



DEALER BEST PRACTICE SERIES

Major Component Tracking System Used
to Monitor & Manage Component Life

Application Maintenance **Site Management** **Component Rebuild** Safety MARC Management

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1.0 Introduction

The Major Component Tracking System (MCTS) is a web based corporate DBS database developed for use by the mining dealers to track major components and component parts.

MCTS:

- Provides a consistent worldwide method for dealers to track major components from cradle-to-grave through all life cycles.
- Provides one safe source for component life data for dealers and Caterpillar.
- Acts as a durability database by providing a complete list of all component lives from new to present.
- Provides actual and complete component life data, which is essential to accurately forecast component life and rebuild frequency when MARC contract costs are being calculated.
- Assists in daily fleet management decisions by providing instant, complete and accurate component history.
- Permits participating dealers to share component life data in the same format.
- Provides numerous, powerful, and user-friendly data analysis tools and reports.
- Use is free of charge to participating dealers. Caterpillar pays for system maintenance and the cost of tools and enhancements.

2.0 Best Practice Description

Type of Data Tracked

MCTS formats component life data by each component life cycle. Each life cycle consists of only a few key pieces of information:

- Component Serial Number
- Installation Date
- Operation Type: (Remove or Install)
- Reason for Removal (Failure, PCR, Wear Out, etc)
- Component Install Type (New, PCR, Dealer Rebuild, Customer Rebuild, ReMan, etc)
- Work Order Number
- Prime Product Serial Number (in which component was installed)
- Prime Product Model Number (793B, 994, etc)
- Prime Product Customer ID (customer fleet number or name used to identify machine)
- Prime Product Hours at Installation or Removal
- Customer Name and Mine Site

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This information is typically obtained from each work order where a tracked major component has been removed or installed.

Create Major Component Operation: Step 2

Header Information:

Serial Number: E480FD102

Manufacturer: CAT

Component Dealer/Customer ID: FD102

Major Component Type: Final Drive

Description:

Scrap Date:

Based on the date entered, Remove is the suggested type for this operation.

Operation Detail:

Main Store Dealer Code: E480 - Wyoming Machinery Company

Operation Date: 27-Mar-2006

Operation Type: Remove

Operation Detail: Select

SMU: Hours

Feet: Gallons

Location On Prime Product: Right Rear

Prime Product SN: 0H0005 select

Prime Product SMU: Hours

Work Order No:

Job Site: Powder River Coal - NARM - Coal

Notes:

Save

Reset

Cancel

MCTS Data Entry Screen
Simple and user-friendly screens provide fast and efficient data entry

Component Serialization

MCTS tracks serialized components in the same way prime products are tracked. Each component must have a unique serial number. Currently, the only major components, which are serialized at the factory, are the engine, transmission, and torque converter. If the component does not have a CAT serial number, a unique identification number may be assigned by the dealer.

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The dealer assigned number is entered into MCTS using the dealer's main store dealer code as a prefix. For example, a *Wyoming Machinery* Final drive is numbered FD102, would be E490FD102 in MCTS. This assures that each component serial number is unique in the world. MCTS will not permit the use of duplicate serial numbers.

MCTS Report Formats:

MCTS provides component life data in three different formats:

1. Component History Report
2. Life Cycle Report
3. Weibull Analysis

Component History Report

The most frequently used report for day-to-day component management is the component history report. In this format, all of the key data for each component installation and removal is located on a single line. All of the installation and removal information for the entire history of a component is listed in chronological order. This makes the data easy to view and understand.

Component History [\[edit header\]](#)

Serial Number: E490FD102 **Manufacturer:** CAT **Component Dealer/Customer ID:** FD102
Component Type: Final Drive **Description:** Cumulative Component SMU: 49,960
Notes: None

[View Current Status](#) [Printable View](#) [Show Critical Parts](#)

Operation History

Oper Date	Oper	Oper Desc	Maj Comp SMU (Hours)	Maj Comp Fuel (Gallons)	Work Order No	Location on Prime Prod	Prime Prod S/N	Prime Prod Sales Mdl	Prime Prod Dir/Cust Id	Prime Prod Mfr	Prime Prod SMU (Hours)	Cust Name	Job Site	Type	Dir
25-Apr-2003	I	Dealer Rebuild	0	0	GL30431	Right Rear	1HL00057	793B	265	CAT	57,897	Powder River Coal	NARM	Coal	E49
01-Apr-2003	R	Failure	3,031	0	GL29924	Right Rear	ATY00435	793C	277	CAT	12,002	Powder River Coal	NARM	Coal	E49
24-Oct-2002	I	Dealer Rebuild	0	0	GL26669	Right Rear	ATY00435	793C	277	CAT	8,971	Powder River Coal	NARM	Coal	E49
01-Oct-2002	R	PCR	10,374	0	GL26164	Right Rear	1HL00140	793B	390	CAT	62,977	Rag Coal West	Eagle Butte	Coal	E49
30-May-2001	I	Dealer Rebuild	0	0	GL14094	Right Rear	1HL00140	793B	390	CAT	52,603	Rag Coal West	Eagle Butte	Coal	E49
29-Apr-2001	R	Failure	90	0	GL13359	Right Rear	4AR00299	793C	221	CAT	21,742	Powder River Coal	NARM	Coal	E49
20-Apr-2001	I	Dealer Rebuild	0	0	GL13171	Right Rear	4AR00299	793C	221	CAT	21,652	Powder River Coal	NARM	Coal	E49
14-Mar-2001	R	PCR	10,998	0	GL12289	Right Rear	1HL00344	793B	220	CAT	35,893	Powder River Coal	NARM	Coal	E49
12-Jul-1999	I	Dealer Rebuild	0	0	433396	Right Rear	1HL00344	793B	220	CAT	24,895	Powder River Coal	NARM	Coal	E49
21-Jun-1999	R	Failure	2,716	0	432977	Right Rear	1HL00392	793B	412	CAT	25,698	Rag Coal West	Belle Ayr	Coal	E49
25-Jan-1999	I	Customer Rebuild	0	0	430359	Right Rear	1HL00392	793B	412	CAT	22,982	Rag Coal West	Belle Ayr	Coal	E49
25-Jan-1999	R	PCR	11,051	0	430359	Left Rear	1HL00392	793B	412	CAT	11,931	Rag Coal West	Belle Ayr	Coal	E49
28-Jul-1997	I	Customer Rebuild	0	0	421873	Left Rear	1HL00392	793B	412	CAT	11,931	Rag Coal West	Belle Ayr	Coal	E49
07-Jul-1997	R	PCR	11,700	0	421542	Right Rear	1HL00329	793B	217	CAT	11,814	Powder River Coal	NARM	Coal	E49
05-Sep-1995	I	New	0	0	412596	Right Rear	1HL00329	793B	217	CAT	114	Powder River Coal	NARM	Coal	E49

Component History Report

Each component life cycle consists of an installation and removal

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Component Life Cycle Report

The life cycle report is the most convenient way to look at how fleet component lives compare to each other in a specific application. Component lives start with “*Life Cycle 1*” (new) and progress chronologically through each component rebuild. This report may also be used to compare the average component life of an older model machine to a newer model in the same application.

Because MCTS provides customer and site-specific data, component lives can be compared from similar applications. However, it should be noted that the sample size declines as life cycles progress. This is because all components have typically completed the early life cycles and the data is mature. Later life cycles list only the components replaced and do not account for those still running in that life cycle.

Life Cycle Summary

Dealer / Group:	E490 - Wyoming Machinery Company	Report Date:	27-Mar-2006
Component Type:	Final Drive	Sales Model Number:	793B
Job Site Type:	All	Job Site:	Powder River Coal - NARM
Life Cycles:	All	Installation Type:	All
Removal Type:	All	Removal Date:	All

[Printable View](#)

Life	Avg. Component Life		Removal Reason							Actual Sample Size
			Failure		Wearout		PCR		Unknown	
	Percent	Avg. Hours	Percent	Avg. Hours	Percent	Avg. Hours	Percent	Avg. Hours	Percent	
1	100	8,258	52%	4,218	0%	0	47%	12,664	1%	46
2	100	9,514	34%	4,466	0%	0	63%	12,001	3%	63
3	100	9,445	38%	6,305	0%	0	61%	11,428	1%	31
4	100	9,751	28%	4,541	0%	0	65%	11,620	7%	32
5	100	8,687	50%	5,403	0%	0	47%	12,152	3%	36
6	100	8,514	50%	5,550	0%	0	50%	11,479	0%	20
7	100	9,001	66%	7,071	0%	0	33%	12,861	1%	9
8	100	8,926	55%	5,845	0%	0	33%	13,172	12%	9
9	100	5,085	75%	3,822	0%	0	25%	8,874	0%	4
10	100	14,200	50%	5,769	0%	0	0%	0	50%	2

MCTS Life Cycle Summary

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Weibull Analysis

Weibull analysis is a statistical tool, which predicts the rate of failure of a population based on the early failures of a few units. This tool has two required inputs:

1. Number of failures and specific hours at which they occurred
2. Number of units still running and hours on those units

Based on this data, Weibull can predict the failure rate of the remainder of the population.

This Weibull tool has been used by Caterpillar for many years, but has been time and labor intensive to use. MCTS has specially designed software to calculate the needed inputs and provide an accurate Weibull analysis in seconds. This is possible because MCTS already contains the number of component replacements (failures) and the hours at which they were replaced. MCTS does not track hours on units running, but does know the installation date, and calculates an average monthly hour accumulation based on the units replaced.

In some cases, one or more early hour component failures due to workmanship may skew the predicted component life downward. These early hour failures may not be indicative of the actual durability of the component population. MCTS has a feature, which allows the early hour failures to be edited out, and the results recalculated. Early hour failures may be edited out in the following increments:

- 0 – 20 hours
- 0 – 200 hours
- 0 – 1,000 hours.

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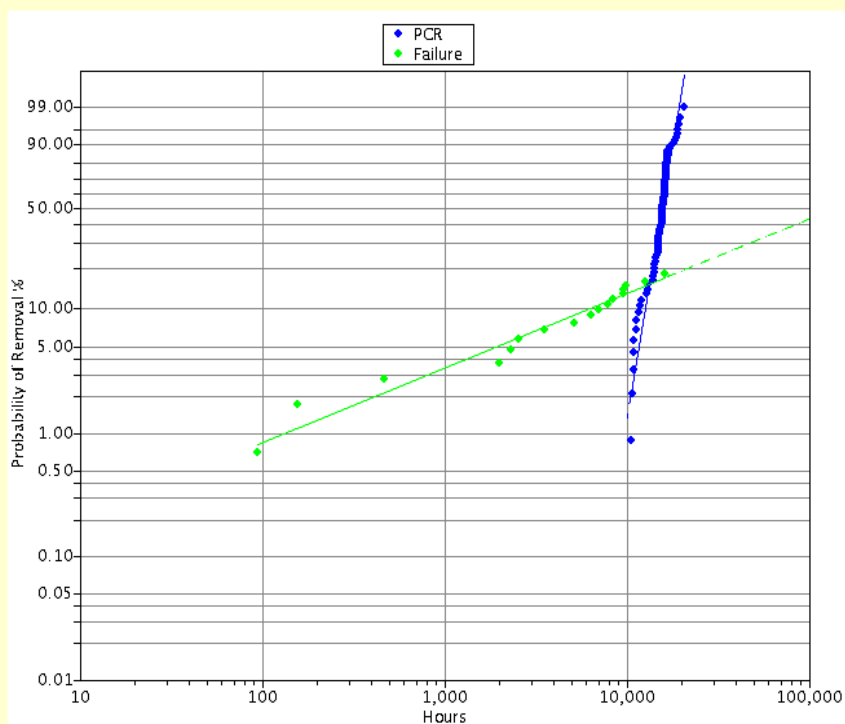
Weibull Analysis

Dealer / Group:	Dealer Code	Report Date:	12-Jan-2006
Component Type:	Engine	Sales Model Number:	793 - All
Job Site Type:	All	Life Cycles:	1st Life
Job Site:	All	Data Point Adjustment:	None

[Printable View](#)

Weibull Analysis Summary

Life	Avg. Component Life		Removal Reason							Actual Sample Size
			Failure		Wearout		PCR		Unknown	
	Percent	Avg. Hours	Percent	Avg. Hours	Percent	Avg. Hours	Percent	Avg. Hours	Percent	
1	100	13,804	16%	6,367	1%	2,005	82%	15,419	1%	98



Thu Jan 12 14:33:28 CST 2006

Note: To be plotted there must be 3 or more entries in the data set. If there are less than 3, the data will show in the Weibull Analysis Summary, but will not be plotted

The Weibull analysis is a powerful tool, which can be used to predict and manage component life proactively.

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3.0 Implementation Steps

MCTS is a powerful tool, which is easy to update and maintain. However, because it tracks component life from new; the complete history of each component tracked must be entered.

The initial data cleansing (removing data sequencing errors & other common data errors), and loading of accurate component history is a one-time exercise, but is absolutely essential. A founding principle of MCTS is that it is better to have no data, than bad data. The amount of labor required to cleanse (remove errors) from the data, is dependent on the number of components being tracked and the quality of dealer component history records. Dealer component history is transferred to a simple Microsoft Excel spreadsheet, which contains all of the required MCTS fields.

A full-time MCTS system administrator is available to help:

- Answer dealer questions
- Provide system user training
- Review cleansed history data for completeness and accuracy before loading

Dealers interested in using MCTS should contact:

Shari Weber
MCTS System Administrator
Weber_Shari_Aat.com
+1 309 675 8115

4.0 Benefits

Implementation and Maintenance Cost

Accurate component life data benefits both Caterpillar and the dealer. The investment for the capability related to component life data cleansing (removing data sequencing errors, mismatches, etc.) and maintaining history data, can be substantial. Caterpillar provides MCTS free of charge. This includes system maintenance and the development of additional tools and enhancements. Once component history has been entered and is current, maintaining the data is very quick and easy. Data entries are only made when a component is removed or installed: typically every 3-4 years.

MCTS Data Access

MCTS is a web-based system that is available to dealers and branch stores anywhere in the world with internet-access. Only authorized and registered users may access MCTS data. Dealers identify individuals they wish to be authorized by name. Caterpillar users also must be registered by name on a need-to-know basis. The MCTS system administrator is the only person who can register individuals and grant access to the system.

Data Storage and Integrity

All component life data is stored on Caterpillar DB2 corporate servers, which are fully protected and maintained by Caterpillar Information Services.

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Ease of Use

MCTS was designed to be very user-friendly and intuitive to use. Most users can navigate the system with little or no training. However, a detailed step-by-step tutorial is also included in the MCTS homepage.

System Speed

The types of standard reports (component history / life cycle / Weibull) provided by MCTS have historically been manually produced and were labor intensive. MCTS uses the power of the DBS servers, along with purpose designed, and proven software to produce the standardized reports in 5 – 10 seconds.

MCTS Capability

Tools and enhancements to the system are not announced until they are fully developed, proven and placed in production. The many features and benefits of MCTS are not promises of future performance. They are fully functional and in use today.

5.0 Resources Required

MCTS is free of charge for Dealers to use. The commitment to enter data and keep it current is a requirement for participating dealers.

A MCTS Dealer Contact person must be appointed for each Dealer. This person is the one point of contact for the MCTS System Administrator for any MCTS related questions and controls Dealer access and entry of data into MCTS. The MCTS Contact person or another designated person or persons, will have the responsibility to enter and maintain the data in MCTS, including the initial data loading process.

The Dealer selects the population to be entered into MCTS. This will usually be the same as what is being tracked today by the dealer's system. If the Dealer desires to only track recently delivered machines, then the Dealer will enter the first change-out information directly into MCTS.

Component Data that has multiple install and remove operations is referred to as "History Data". If Dealer desires to enter this data, it must be transferred to the MCTS Data Input Microsoft Excel Spreadsheet, and submitted to the MCTS System Administrator. Data is checked and corrected in this process for sequence errors and other common data errors. Once the data is approved by the Dealer and the MCTS System Administrator, the data is uploaded into MCTS.

The Manpower Requirements for this History Data loading process varies by Dealer depending on the quantity and quality of the data, it can take from 1-3 man months.

6.0 Supporting Attachments / References

- *MCTS Intro - Microsoft PowerPoint file* located in the Attachments Tab of this pdf document.
- Other documents can be found on Catminer.com

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7.0 Related Best Practices

None applicable

8.0 Acknowledgements

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